SPURVANCE

Cyber Security Course

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Assignment #01

My Digital Life Map: Understanding Personal Data, Storage Locations, and Cybersecurity Risks

Objective:

The objective of this assignment is to identify the digital applications and accounts used in daily life, understand the types of personal data they store, and learn where this data is stored. It also aims to identify high-risk applications and explain the importance of cybersecurity and the CIA Triad in protecting personal and sensitive information.

Five Daily Apps or Accounts:

The following five applications are commonly used for communication, education, financial transactions, and social networking.

- **WhatsApp:**

Purpose: WhatsApp is used for messaging, voice and video calls, and sharing photos, videos, documents, and voice notes.

Data Stored: Chat messages, media files, contacts, profile information, and voice notes.

Storage Location: Messages and media may be stored on the device and, depending on settings, in cloud backups. WhatsApp also processes certain account and service-related information on its servers.

Security Importance: Two-step verification should be enabled, and backup security settings should be reviewed.



- **Gmail:**

Purpose: Gmail is an email service used for sending and receiving messages, documents, and attachments.

Data Stored: Emails, attachments, contacts, account information, and email settings.

Storage Location: Emails are primarily stored on Google's cloud servers. Some information may also be cached locally on the device.

Security Importance: A strong password and two-factor authentication help protect the account from unauthorized access.



- **JazzCash:**

Purpose: JazzCash is a mobile financial service used for money transfers, bill payments, mobile top-ups, and other financial transactions.

Data Stored: Transaction history, wallet information, account details, and identity-verification information, depending on the account and service used.

Storage Location: Financial and account records are processed and stored in the service provider's systems. Some information may also appear in the application on the device.

Security Importance: The account PIN and verification codes must remain private. Unrecognized transactions should be reported immediately.



- **Instagram:**

Purpose: Instagram is a social media application used for sharing photos, videos, stories, reels, and direct messages.

Data Stored: Profile information, uploaded media, direct messages, followers, following lists, and account activity.

Storage Location: Most account information and uploaded content are stored or processed on Instagram's cloud infrastructure. Some data may be cached on the device.

Security Importance: A private account, strong password, two-factor authentication, and careful review of connected applications can improve security.



- **University Portal:**

Purpose: A university portal allows students to access academic information and educational services.

Data Stored: Student details, grades, course registrations, attendance records, fee vouchers, and academic information, depending on the portal.

Storage Location: Information is generally stored and managed on university servers or the cloud services used by the institution.

Security Importance: Students should use strong passwords, log out from shared computers, and avoid sharing login credentials.



Data Types and Storage Locations:

Digital applications collect and manage different types of data. The following table summarizes the information associated with the five selected applications.

Storage arrangements may vary depending on application settings, backup configuration, and the service provider.

High-Risk Application: JazzCash

JazzCash has been selected as the high-risk application because it is connected to financial transactions and may contain sensitive personal and identity information.

If an unauthorized person gains access to a JazzCash account, they may attempt fraudulent transactions, misuse account information, or access sensitive financial records. This can cause financial loss, privacy violations, and inconvenience for the account holder.

CIA Triad Impact:

The CIA Triad is a cybersecurity model consisting of Confidentiality, Integrity, and Availability. It helps explain the main goals of information security.

Confidentiality:

Confidentiality means ensuring that information is accessible only to authorized people. If a JazzCash account is compromised, an attacker may view private transaction details or sensitive account information.

Integrity:

Integrity means maintaining the accuracy and reliability of information. An attacker may attempt unauthorized transactions or manipulate financial records, affecting their accuracy.

Availability:

Availability means ensuring that systems and information remain accessible when needed. An account compromise, service disruption, or security-related account restriction may prevent a legitimate user from accessing financial services temporarily.

Security Recommendations:

- Use a strong, unique password wherever applicable.
- Keep the account PIN and one-time verification codes confidential.
- Never click suspicious links or respond to fraudulent messages.
- Use the official application and keep it updated.
- Monitor transaction notifications and report suspicious activity promptly.
- Never share sensitive account screenshots in public reports.

Definition

Cybersecurity

Cybersecurity is the practice of protecting computers, networks, applications, and digital information from unauthorized access, attacks, damage, and theft.

Data

Data refers to information stored in digital form, such as text, numbers, photographs, videos, and audio files. Applications process data to provide useful services.

Software

Software is a collection of programs and instructions that tell a computer or mobile device what to do. Examples include WhatsApp, Gmail, and operating systems.

Server

A server is a computer or system that provides data, resources, or services to other computers and devices over a network.

Cloud

The cloud refers to remote computing resources accessed through a network, usually the internet. It allows users and organizations to store, manage, and process information without relying entirely on a local device.

Conclusion

This assignment helped explain how everyday digital applications collect, store, and process different types of personal information. WhatsApp, Gmail, JazzCash, Instagram, and university portals serve different purposes, but all require appropriate security measures.

The analysis identified JazzCash as a high-risk application because unauthorized access may affect financial privacy, transaction integrity, and access to financial services. Understanding the CIA Triad helps users recognize these risks and apply suitable protective measures.

Overall, using strong passwords, enabling available security features, protecting verification codes, and avoiding suspicious links are important steps toward maintaining digital privacy and security.